

Bach Do

Engineering Building 2, Room E121
4226 Martin Luther King Boulevard
Houston, TX 77204

E-mail: dokimbach@gmail.com | Phone number: (346) 397-8686

PROFESSIONAL APPOINTMENT

University of Houston Postdoctoral fellow	Aug. 2023 –
Kyoto University Student research assistant	Jul. 2020 – Mar. 2023
GBC Engineers GmbH Senior structural engineer	Sep. 2018 – Aug. 2019
CC1-JSC Civil engineer	Jan. 2015 – Jun. 2016

EDUCATION

Kyoto University Ph.D. Architectural Engineering Dissertation: Probabilistic approaches to optimization of steel structures considering uncertainty. (Link)	Mar. 2023
Kyoto University Research student Architectural Engineering.	Oct. 2019–Mar. 2020
Chulalongkorn University M.Eng. Structural Engineering Thesis: Optimum patch design for repairing cracked steel plates using genetic algorithm. (Link)	Aug. 2018
Ho Chi Minh City University of Technology B.Eng. (Honors) Civil Engineering	Apr. 2015

RESEARCH INTEREST

- Bayesian optimization
- Probabilistic machine learning
- Multi-fidelity modeling
- Uncertainty quantification

I am interested in developing reliable and interpretable methods for engineering optimization, data-driven engineering, and uncertainty quantification, with a focus on applications in structural and mechanical systems.

AWARDS AND HONORS

Young Encouragement Award Kyoto University	Sep. 2022
Recipient of Scholarships for Graduate Studies The Japan International Cooperation Agency (JICA)	2016–2018, 2019–2023

Updated January 31, 2026

PUBLICATIONS

Articles

1. **Sampling from Gaussian processes: A tutorial and applications in global sensitivity analysis and optimization.**
Bach Do, Nafeezat Ajenifuja, Taiwo Adebiyi, Ruda Zhang.
Structural and Multidisciplinary Optimization. ([pdf](#))
2. **Multi-fidelity Bayesian optimization: A review.**
Bach Do, Ruda Zhang.
AIAA Journal (2025). ([Link](#))
3. **Optimizing posterior samples for Bayesian optimization via rootfinding.**
Taiwo Adebiyi, Bach Do, Ruda Zhang.
The Thirteenth International Conference on Learning Representations (ICLR 2025). ([OpenReview](#)), ([Poster](#)), ([Python package](#))
4. **Automated design of nonreciprocal thermal emitters via Bayesian optimization.**
Bach Do, Sina J. Ghalekohneh, Taiwo Adebiyi, Bo Zhao, Ruda Zhang.
Journal of Quantitative Spectroscopy and Radiative Transfer (2025). ([Link](#))
5. **Combined surrogate-assisted Bayesian optimization and comprehensive learning PSO method for parameter identification of ultra-high strength circular CFST short columns.**
Cuong Pham Van Le, Sawekchai Tangaramvong, Thu Van Huynh, Bach Do, Quang-Viet Vu, Wei Gao.
Probabilistic Engineering Mechanics (2025). ([Link](#))
6. **Gaussian process Thompson sampling via rootfinding.**
Taiwo Adebiyi, Bach Do, Ruda Zhang.
NeurIPS 2024 Workshop on Bayesian Decision-making and Uncertainty, 2024 (top 5% of accepted papers). ([Link](#))
7. **Epsilon-greedy Thompson sampling to Bayesian optimization.**
Bach Do, Taiwo Adebiyi, Ruda Zhang.
Journal of Computing and Information Science in Engineering (2024). ([Link](#))
8. **A novel decoupled approach combining invertible cross-entropy method with Gaussian process modeling for reliability-based design and topology optimization.**
Thu Van Huynh, Sawekchai Tangaramvong, Bach Do, Wei Gao.
Computer Methods in Applied Mechanics and Engineering (2024). ([Link](#))
9. **Bayesian optimization-assisted approximate Bayesian computation and its application to identifying cyclic constitutive law of structural steels.**
Bach Do, Makoto Ohsaki.
Computers & Structures (2023). ([Link](#))
10. **Sequential most probable point update combining Gaussian process and comprehensive learning PSO for structural reliability-based design optimization.**
Thu Van Huynh, Sawekchai Tangaramvong, Bach Do, Wei Gao, Suchart Limkatanyu.
Reliability Engineering & System Safety (2023). ([Link](#))
11. **Two-step parameter identification of multi-axial cyclic constitutive law of structural steels from cyclic structural responses.**
Makoto Ohsaki, Bach Do, Jun Fujiwara, Toshiaki Kimura, Takuzo Yamashita.
Structures (2022). ([Link](#))

12. **Proximal-exploration multi-objective Bayesian optimization for inverse identification of cyclic constitutive law of structural steels.**
Bach Do, Makoto Ohsaki.
 Structural and Multidisciplinary Optimization (2022). ([Link](#))
13. **Bayesian optimization for inverse identification of cyclic constitutive law of structural steels from cyclic structural tests.**
Bach Do, Makoto Ohsaki.
 Structures (2022). ([Link](#))
14. **Sequential sampling approach to energy-based multi-objective design optimization of steel frames with correlated random parameters.**
Bach Do, Makoto Ohsaki.
 Earthquake Engineering & Structural Dynamics (2022). ([Link](#))
15. **Bayesian optimization for robust design of steel frames with joint and individual probabilistic constraints.**
Bach Do, Makoto Ohsaki, Makoto Yamakawa.
 Engineering Structures (2021). ([Link](#))
16. **Sequential mixture of Gaussian processes and saddlepoint approximation for reliability-based design optimization of structures.**
Bach Do, Makoto Ohsaki, Makoto Yamakawa.
 Structural and Multidisciplinary Optimization (2021). ([Link](#))
17. **A random search for discrete robust design optimization of linear-elastic steel frames under interval parametric uncertainty.**
Bach Do, Makoto Ohsaki.
 Computers & Structures (2021). ([Link](#))
18. **Gaussian mixture model for robust design optimization of planar steel frames.**
Bach Do, Makoto Ohsaki.
 Structural and Multidisciplinary Optimization (2021). ([Link](#))
19. **Optimization of fiber-reinforced polymer patches for repairing fatigue cracks in steel plates using a genetic algorithm.**
Bach Do, Akhrawat Lenwari.
 Journal of Composites for Construction (ASCE) (2020). ([Link](#))

Book Chapters

1. **Design optimization of adhesive-bonded FRP patches for repairing fatigue cracks in steel structures (Chapter 10).**
 Akhrawat Lenwari, Bach Do.
 In Vistasp M. Karbhari (Editor), Rehabilitation of Metallic Structural Systems Using Fiber-Reinforced Polymer (FRP) Composites, 2nd Edition, Elsevier (2025). ([Link](#))

PRESENTATIONS

1. **Multi-fidelity Bayesian optimization in engineering design.**
Bach Do, Ruda Zhang.
 The Engineering Mechanics Institute Conference and Probabilistic Mechanics & Reliability Conference (EMI/PMC 2024), Chicago, Illinois (2024).
2. **Bayesian optimization for inverse identification of cyclic constitutive law of structural steels.**

Bach Do, Makoto Ohsaki.

The 15th World Congress on Computational Mechanics & 8th Asian Pacific Congress on Computational Mechanics (WCCM-APCOM 2022), Yokohama, Japan (2022).

3. Inverse identification of cyclic constitutive law of structural steels using multi-objective Bayesian optimization.

Bach Do, Makoto Ohsaki.

The 17th East Asia-Pacific Conference on Structural Engineering and Construction (EASEC-17), Singapore (2022).

4. Sequential batch sampling approach to discrete multi-objective reliability-based design optimization.

Bach Do, Makoto Ohsaki.

The Asian Congress of Structural and Multidisciplinary Optimization 2022 (ACSMO2022), Matsue, Japan (2022).

5. A novel saddlepoint approximation and its application to structural reliability-based design optimization.

Bach Do, Makoto Ohsaki.

The 14th World Congress of Structural and Multidisciplinary Optimization (WCSMO14), Boulder, Colorado, USA (2021).

6. A Gaussian mixture model for multi-objective robust design optimization of steel frames under aleatory uncertainty.

Bach Do, Makoto Ohsaki.

The 3rd International Conference on Computational Engineering and Science for Safety and Environmental Problems (COMPSAFE2020), Kobe, Japan (2020).

7. A novel saddlepoint approximation and its application to structural reliability-based design optimization.

Bach Do, Makoto Ohsaki.

Annual meeting (Structure I), Architectural Institute of Japan (AIJ), Kanto, Japan (2020).

TEACHING EXPERIENCE

Introduction to Numerical Optimization

Fall, 2023

Uncertainty Quantification Lab at University of Houston

ACADEMIC SERVICE

Journal Review

- Applied Mathematical Modelling
- Structural and Multidisciplinary Optimization
- Computer Methods in Applied Mechanics and Engineering
- Reliability Engineering & System Safety
- Engineering Structures
- Aerospace Science and Technology
- Expert Systems with Applications

Mentoring PhD Students

- Taiwo Adebisi (University of Houston)
- Nafeezat Ajenifuja (University of Houston)

ACCADEMIC AND PROFESSIONAL AFFILIATIONS

- American Society of Civil Engineers (ASCE)
- International Society for Structural and Multidisciplinary Optimization (ISSMO)

REFERENCES

Prof. Makoto Ohsaki (Ph.D. advisor)

Department of Architecture and Architectural Engineering, Kyoto University

Address: Kyotodaigaku-Katsura, Nishikyo-ku, Kyoto 615-8540

E-mail: ohsaki@archi.kyoto-u.ac.jp

Phone: (+81) 75-383-2923

Dr. Ruda Zhang (Postdoc host)

Department of Civil & Environmental Engineering, University of Houston

Address: 4226 Martin Luther King Boulevard, Houston, TX 77204-4003

E-mail: rudaz@uh.edu

Phone: (713) 743-0681

Prof. Akhrawat Lenwari (Master advisor)

Department of Civil Engineering, Chulalongkorn University

Address: 254 Phayathai Rd, Khet Pathum Wan, Bangkok 10330

E-mail: akhrawat.l@chula.ac.th

Phone: (+66) 22-186-693

Dr. Bo Zhao (Collaborator)

Department of Mechanical and Aerospace Engineering, University of Houston

Address: 4226 Martin Luther King Boulevard, Houston, TX 77204-4003

E-mail: bzhao8@uh.edu

Phone: (713) 743-2557